Toward Generative Design of Aircraft-Like Structures with Diffusion Models and CFD-Informed Evaluation

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***Abstract*—This paper documents a proof-of-concept generative-design pipeline for aircraft-like voxel geometries. The repository combines a latent diffusion-style model, a voxel decoder, connectivity heuristics, and a CFD-informed scoring path built around an internal lattice-Boltzmann solver plus an external OpenFOAM export route. The present implementation is intentionally scoped: training data are synthetic, the reported experiment is a reduced sanity run, and the evidence should be read as code-path validation rather than publication-grade aerodynamic or structural validation. In addition to describing the architecture and benchmark workflow, we use the current experiments to clarify which claims the codebase supports today and which claims, including mission-conditioned or manufacturing-conditioned aircraft generation, remain future work.**

***Index Terms*—Generative Design, Diffusion Models, Computational Fluid Dynamics, Aerospace Engineering, Structural Optimization.**

I. Introduction

GENERATIVE design is increasingly used to explore large engineering design spaces, while aerospace workflows continue to rely heavily on computational fluid dynamics (CFD), surrogate models, and optimization loops to evaluate aerodynamic tradeoffs [1], [2]. Aircraft design adds further constraints beyond raw shape generation, including manufacturability, structural load paths, and mission-specific performance targets. Those requirements make it easy to overstate what a proof-of-concept generative repository can presently demonstrate.

This paper therefore takes a deliberately narrower position. The repository implements a latent diffusion-style pipeline that maps latent samples to voxel geometries, scores them with an internal CFD-oriented path, and exports them for downstream validation. The present evidence comes from synthetic training data and reduced sanity experiments, so the goal of this paper is not to claim a finished aircraft-design system. Instead, the goal is to document the implementation, report what the current code path does reproduce, and make the boundary between demonstrated behavior and future work explicit.

The key contributions of this work are:

1) A reproducible proof-of-concept latent generative pipeline for producing aircraft-like voxel geometries and exporting them to mesh-based artifacts.
2) An implementation path for CFD-informed evaluation that couples an internal lattice-Boltzmann-based scorer with an external OpenFOAM benchmark route, together with a reduced sanity experiment showing that the end-to-end path executes.
3) An issue-driven validation framework that distinguishes code-path validation from stronger claims such as aircraft-specific validity, aerodynamic optimization, structural viability, and conditioned generation from mission or manufacturing constraints.

The rest of the paper is organized as follows. Section II positions the repository relative to 3D generative modeling, topology optimization, and CFD-driven design practice. Section III describes the implementation that currently exists in the codebase. Section IV reports the reduced sanity-run evidence and explains its limits. Section V records the validation requirements that remain before stronger claims can be made. Section VI summarizes the present scope and the minimum future work needed for a conditioned airplane generator.

II. Related Work

Diffusion models are now a standard point of reference for high-quality generative modeling [3], [4]. In 3D generation, recent systems such as Point-E and Shap-E show that diffusion-style methods can generate point-cloud or implicit 3D assets at practical sampling speeds [5], [6]. Those works are important context because they establish that diffusion models can represent 3D geometry, but they do not by themselves solve aircraft-specific design, manufacturability, or validation concerns.

An adjacent engineering-design tradition comes from topology optimization, where the goal is to optimize material layout or geometry under physics-based objectives and constraints. That literature is mature and has a well-developed vocabulary for separating optimization variables, constraints, and validation requirements [7]. Our repository is closer to a generative prior over voxelized shapes than to classical topology optimization, but topology-optimization practice remains a useful baseline for how strongly structural or performance claims should be validated.

Conventional aerodynamic shape optimization couples CFD solvers with numerical optimization over parameterized geometries, often using gradient or adjoint information [1]. In that setting, the geometry representation, the optimizer, and the CFD solver are all explicit components of a tightly controlled loop. By contrast, the present repository samples

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from a learned latent model and only then applies CFD-style evaluation. This makes the workflow exploratory, but it also means that success in code-path execution should not be confused with the stronger evidence usually expected in mature CFD-based shape optimization.

Recent surrogate-model and differentiable-simulation work aims to reduce the cost of repeated flow evaluation, either by learning fast predictors of aerodynamic coefficients [2] or by embedding differentiable fluid solvers into learning and design frameworks [8]. These directions are especially relevant because they represent stronger versions of "simulation in the loop" than the reduced sanity evidence reported here.

The current implementation also borrows standard architectural ingredients from modern deep learning, including attention-style modules inspired by the Transformer literature [**?**]. In this paper, those ingredients are treated as reused components rather than as a novelty claim.

A. Positioning and Novelty

The present repository does not claim a new diffusion objective, a new adjoint CFD method, or a validated aircraft-design benchmark. Its most defensible novelty is narrower: a proof-of-concept assembly of a latent generative model, voxel decoding path, CFD-informed evaluation hook, STL export, and benchmark discipline in a single reproducible codebase. Relative to general 3D diffusion work, the emphasis here is on engineering-shaped voxel outputs rather than text-to-3D asset generation. Relative to topology optimization and classical CFD shape optimization, the repository explores a learned generative prior rather than directly optimizing a single parameterized shape. Relative to surrogate and differentiable-simulation papers, the current implementation is lighter-weight and presently supported only by sanity-run evidence. Mission-conditioned generation, manufacturing-conditioned generation, aircraft-specific validity checks, and publication-grade aerodynamic validation remain future work.

B. Generative Design and Topology Optimization

Topology Optimization (TO), specifically the SIMP method [9], is the industry standard for load-path optimization [10]. Recent neural TO approaches [11] accelerate this process. However, TO is often sensitive to local minima and initial conditions. A future comparative study could test whether a learned generative prior explores shape families that differ from those reached by gradient-based material redistribution alone.

C. Aerospace and Aerodynamic Shape Optimization

Modern aerodynamic shape optimization (ASO) relies on adjoint methods for gradient calculation [12], [13] and surrogate modeling for efficient design space exploration [14]. Our framework uses an integrated D3Q27 LBM solver to provide simulation-in-the-loop 3D aerodynamic feedback, positioning it as an exploratory tool that complements local adjoint-based optimization. The preliminary accuracy of this feedback is documented in Section IV.

D. Physics-Informed and Simulation-in-the-Loop ML

Physics-informed machine learning incorporates PDEs through PINNs [15] or Neural Operators like FNO [16] and DeepONet [17]. Our work follows a simulation-in-the-loop paradigm where an internal LBM-style solver provides bounded guidance during training [18]. A stronger future version could use that path for multi-fidelity label promotion from the internal solver to higher-fidelity external CFD, but that workflow is not yet a validated result in this paper.

E. Constraint-Aware Generation

In practical terms, the current repository evidence remains limited to connectivity penalties, post-processing cleanup, and manufacturing-aware scoring proxies. These mechanisms help smoke-test the code path, but they should not be read as aircraft-specific validity guarantees. Enforcing physical validity—such as manufacturability and connectivity—remains a core challenge [19]. Oh et al. [20] demonstrated generative design for structures, while Steinmetz et al. [21] utilized differentiable projection for manufacturing. Our implementation extends this through a persistent Constraint Projector and semantic repair adapter designed to ensure designs comply with aircraft-specific clearance and connectivity rules. The effectiveness of these constraints is partially probed by our planned cleanup-threshold sweep in Section IV.

F. Novelty and Positioning

To keep the claim boundary honest, the narrower interpretation in this revision is that the repository assembles a reproducible proof-of-concept workflow and not that it has already demonstrated aircraft-specific validity, superior exploration, or validated surrogate replacement.

This work is distinguished from prior art as follows:

- **vs. 3D Diffusion:** The repository focuses on voxelized engineering-shaped artifacts and code-path validation rather than on general-purpose 3D asset generation.
- **vs. Topology Optimization:** The current workflow explores a learned generative prior rather than directly optimizing a single parameterized geometry.
- **vs. CFD Surrogates:** The current branch keeps a direct internal scoring hook in the loop instead of replacing it with a fixed learned surrogate.

This positioning is intentionally narrow and is supported only by the implementation and smoke evidence documented in Section IV.

G. Experimental Baseline Selection

The items below are a planned evaluation protocol for future claim expansion. They are not completed experiments in the current paper revision.

To rigorously evaluate the proposed framework, we define a planned set of baseline comparisons and ablation studies. They are included here as a future evaluation protocol, not as completed experiments in the current paper revision.

1) Baseline Families: We compare our method against four primary families of baselines, representing the state of the art in traditional and data-driven design:

1) **Classical Primitives:** We utilize a library of hand-designed parametric aircraft primitives, specifically swept and tapered wings extruded from NACA 0012 and 4412 airfoil profiles. Parameters include aspect ratio (4–12), sweep angle (0–45°), and taper ratio (0.4–1.0). These represent the "gold standard" of traditional aero-structural design.
2) **Unconstrained Generative 3D:** We will evaluate against representative models for voxel, point-cloud, and implicit representations, specifically voxel-GAN [22], point-cloud diffusion [23], and implicit occupancy networks [24]. These models are trained on the same synthetic dataset but lack our aerodynamic guidance and connectivity priors, allowing us to isolate the impact of our constraints.
3) **Independent Optimization Search:** To benchmark the efficiency of the diffusion process, we compare against (a) a SIMP-based topology optimization [9] for material distribution, and (b) an independent random-search baseline where 1,000 design candidates are sampled and ranked directly by high-fidelity D3Q27 CFD simulations.
4) **Differentiable Aerodynamic Shape Optimization (ASO):** We include a baseline using adjoint-based gradient descent [13] to refine the mesh of a standard fuselage-wing primitive, testing whether generative exploration can find superior global optima compared to local refinement.

2) Ablation Studies: We investigate the contribution of individual components through a planned experimental protocol:

- **Physics Ablation:** Training without the $\mathcal{L}_{aero}$ term to evaluate the specific impact of CFD guidance on the resulting L/D distribution.
- **Structural Ablation Suite:** To isolate the effects of our manifold integrity mechanisms, we compare (a) a "no-constraint" model (no $\mathcal{L}_{conn}$), (b) a "no-cleanup" model (disabling the connected-component post-processing hook), and (c) a "no-projection" model (disabling the semantic 'ConstraintProjector').
- **Consistency Step Sweep:** Evaluating the Pareto frontier of design quality versus inference latency across step counts $N \in \{1, 2, 4, 8, 16\}$.

3) Claim-to-Baseline Mapping: Table I maps our primary research claims to specific baselines, metrics, and execution specifications.

III. METHODOLOGY

The current repository implements a proof-of-concept generative workflow built from four components: a noise scheduler, a latent diffusion-style denoiser, a latent-to-3D converter, and an internal CFD-oriented evaluator. The codebase operates on synthetic voxel training data and uses bounded smoke runs together with connectivity heuristics and aerodynamic score terms. In this revision, the same path also carries a documented structured condition vector through dataset generation, latent construction, and checkpointed inference. Figure 1 should therefore be read as an implementation diagram for the present repository rather than as evidence of a fully validated aircraft-design system.

A. Noise Scheduling

The forward diffusion process gradually adds Gaussian noise to the input data over a series of T timesteps. The noise schedule is defined by a variance schedule β_t, which is typically a linear or cosine schedule. In our work, we use a linear schedule, where β_t increases linearly from β_{start} to β_{end}.

The forward process is defined as:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \quad (1)$$

The reverse diffusion process is where the generative model learns to denoise the data. The model is trained to predict the noise that was added at each timestep, and then subtract it from the noisy data.

B. Latent Diffusion UNet

The core of our generative model is a UNet-based architecture that operates in the latent space. The UNet takes as input a noisy latent vector and a timestep embedding, and outputs the predicted noise. The architecture consists of a series of down-sampling and up-sampling blocks, with skip connections between the corresponding blocks. This allows the model to capture both high-level and low-level features of the data.

C. Latent-to-3D Converter

The latent-to-3D converter is a multi-layer perceptron (MLP) that maps the denoised latent vector to a 3D voxel grid. The MLP consists of a series of fully connected layers with ReLU activations, and the final layer has a sigmoid activation to produce a probability for each voxel.

D. CFD Simulator

The repository uses an internal lattice-Boltzmann-style CFD evaluator to score generated voxel grids. The evaluator takes a voxel geometry and flow settings such as Mach number or Reynolds number and returns drag- and lift-related quantities. In addition, the repository can export benchmark cases for comparison against an external OpenFOAM run. This combination is useful for code-path validation, but the present paper does not treat it as a substitute for a full aerodynamic validation campaign.

E. Loss Function

The total loss function is a weighted sum of three components: the diffusion loss, the connectivity loss, and the aerodynamic loss.

$$\mathcal{L} = \lambda_{diff}\mathcal{L}_{diff} + \lambda_{conn}\mathcal{L}_{conn} + \lambda_{aero}\mathcal{L}_{aero} \quad (2)$$

TABLE I
EXPERIMENTAL PLAN AND CLAIM MAPPING

Research Claim	**Baseline / Ablation**	**Evaluation Metrics**	**Planned Protocol**
Aerodynamic efficiency	Classical Primitives; Adjoint ASO; No-CFD-loss	C_L/C_D; Drag count ($C_D \times 10^4$)	1,000-step D3Q27 at 128^3 res; 50 samples per configuration.
Manifold integrity	Voxel-GAN; Point-cloud Diffusion; No-cleanup	Connected component ratio (%); Repair success rate (%)	Voxel-connectivity analysis of 200 random designs at 32^3 res.
Computational efficiency	Random-search + CFD ranking; DDPM (1,000 steps)	Inference latency (s); VRAM (GB); L/D convergence rate	Wall-clock timing on RTX 3090/A100; batch size 4; 100 samples.
Generative diversity	Parametric Primitives (swept/tapered wing library)	MS-SSIM; Latent-space coverage; Parts-volume variance	Conditioned batch gen (10 designs $\times$ 6 classes $\times$ 5 missions).

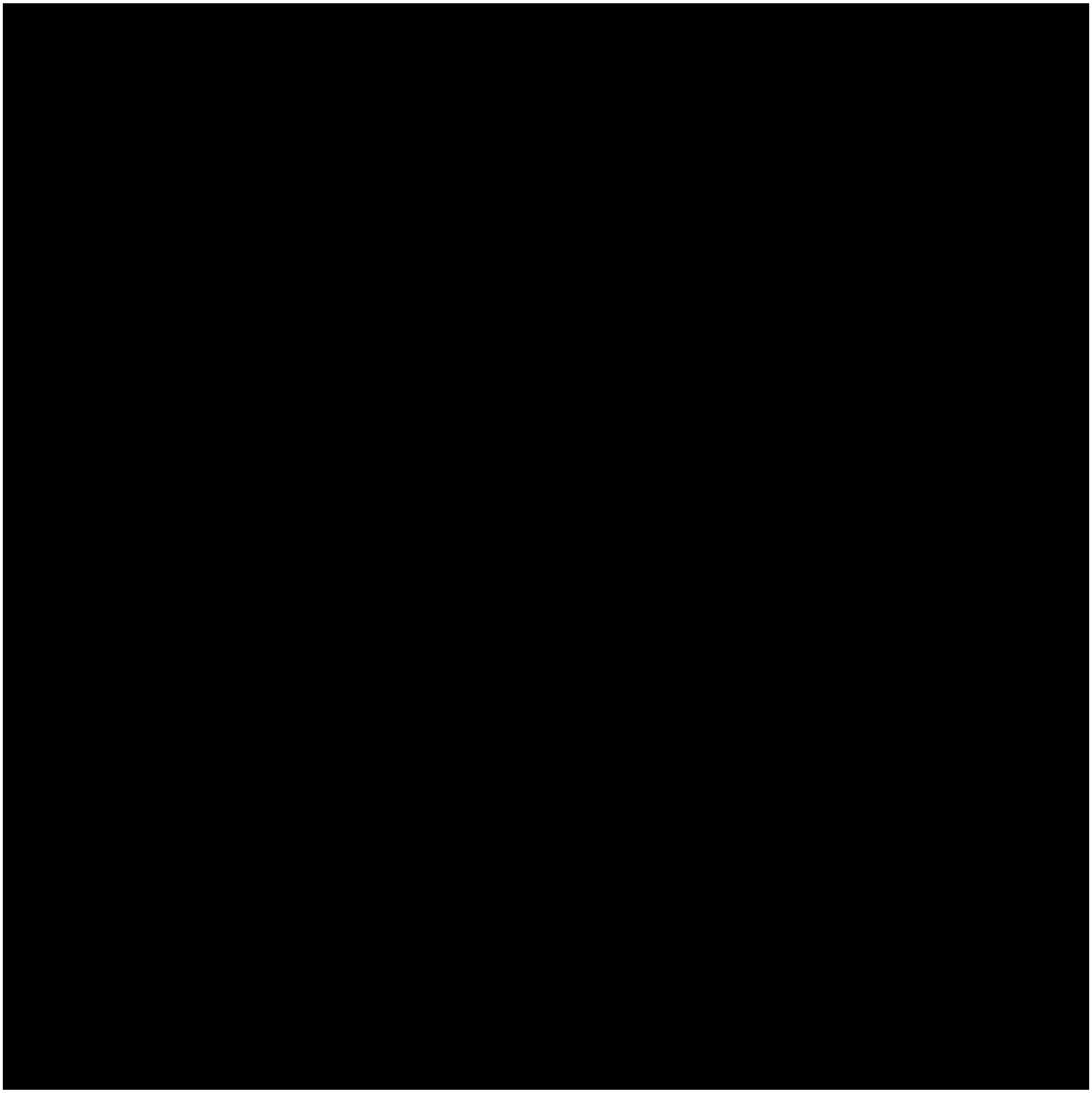

Fig. 1. Implementation-level view of the current repository. A latent code is sampled, denoised by the latent model, decoded into a voxel grid, and then evaluated through internal scoring and export paths. The figure represents the intended code path; it does not by itself establish differentiable or publication-grade CFD validation.

The diffusion loss is the mean squared error between the predicted noise and the actual noise. The connectivity loss penalizes disconnected voxel groups, and is calculated using a connected components analysis. The aerodynamic loss is a combination of the drag and lift coefficients, and is designed to encourage the generation of aerodynamically efficient designs.

F. Current CFD Coupling In The Training Loop

The codebase is designed to couple geometry generation and CFD-informed scoring during training. In the present implementation, that coupling should be understood as a practical score path rather than as a demonstrated differentiable CFD-training result. The current repository evidence is limited to reduced sanity runs, so we only claim that the training code can invoke aerodynamic scoring terms and route generated geometries through the evaluator.

- **Step 1:** Sample noisy latent z_t.
- **Step 2:** Predict z_0 using the UNet.
- **Step 3:** Decode z_0 to voxel grid V.
- **Step 4:** Run GPU-accelerated LBM solver on V.
- **Step 5:** Compute aerodynamic coefficients C_L, C_D.
- **Step 6:** Update model weights using the resulting scalar loss terms.

In a stronger future version of this pipeline, the repository would need ablations showing that the aerodynamic term measurably changes either the training dynamics or the ranking of generated candidates. Until such evidence exists, the paper treats the CFD term as an implemented mechanism whose effect is not yet fully validated.

IV. Results and Discussion

The empirical evidence in this revision is intentionally narrow. It consists of bounded smoke runs on synthetic or procedural data, a reduced freeform-object sweep for the historical CFD path, and smoke-level checks for the new structured-conditioning seam. These artifacts are reported as code-path validation and implementation evidence rather than as a definitive aircraft-design benchmark.

A. Training Convergence

Figure 2 summarizes a historical reduced training trace retained for context. The main use of this figure is to show that the implementation can execute the loss path end to end on a bounded smoke run. It should not be read as a convergence study or as evidence that the current repository has validated aerodynamic or structural learning behavior.

B. Freeform-object Sweep: Steps vs. Shape Prior

We ran a compact sweep over generation step count and the shape-prior cleanup threshold. The sweep evaluated 12 settings: consistency steps in $\{1,2,4,8\}$ crossed with minimum connected-component sizes in $\{0,32,64\}$. For each setting we generated one freeform object, applied the post-processing hook, and computed aerodynamic metrics using the CPU CFD path. The results in Figure 3 provide a preliminary probe of how post-processing thresholds change the resulting aerodynamic score distribution.

Fig. 2. Observed historical smoke-run losses from an earlier reduced training trace. The figure is retained as implementation evidence only, not as a full convergence study.

Fig. 3. Mean L/D from the reduced sweep over consistency steps, aggregated across three post-processing thresholds. Error bars show sample standard deviation.

C. Freeform-object Geometry Summary

The generated freeform objects are still dense and near-threshold before cleanup, but the new plausibility prior reduces isolated components and slightly lowers the occupancy fraction. In this environment, that matters more than small changes in the raw voxel logits: the cleaned binary shapes are less

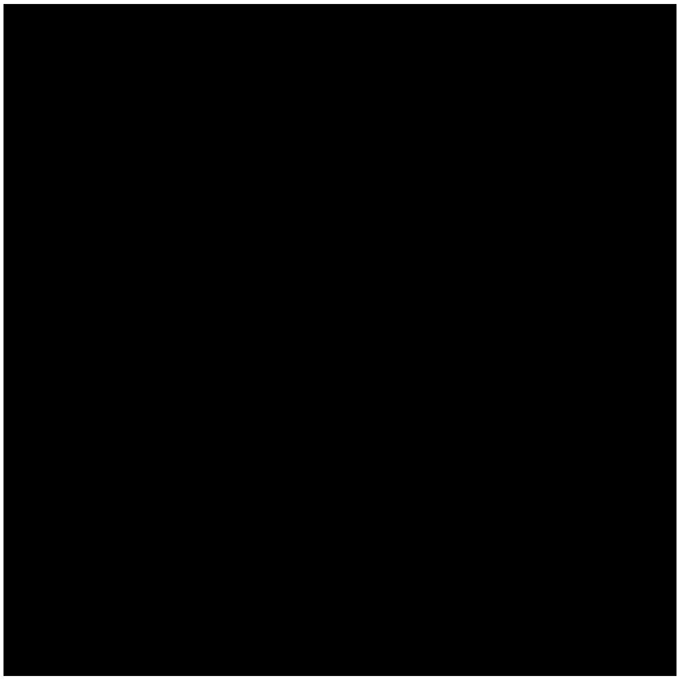

Fig. 4. Occupancy versus L/D for the same sweep. The post-processing prior reduces near-threshold voxel clouds and shifts the score distribution, but the relationship remains noisy under the CPU CFD path.

fragmented and more suitable for downstream meshing and export. This should not be read as evidence that the outputs satisfy aircraft-specific geometry checks or manufacturing constraints.

D. Hyperparameter and Tuning Notes

The sweep was intentionally small but meaningful for the current codebase: it varied the inference path length and the new connected-component cleanup threshold. The training-side improvement was also incremental: a voxel-shape prior was added to the loss to discourage midpoint noise and disconnected fragments. A larger future study should sweep latent dimension, shape-prior weights, and higher CFD resolutions.

E. Structured-Conditioning Smoke Evidence

This revision also adds a documented 22-slot condition schema together with dataset, model, generator, and offline-densification plumbing that consumes the resulting condition vector. We include a smoke-level condition-response summary on the procedural path to verify that materially different condition payloads produce measurable directional deltas in simple geometry proxies such as occupancy, span, and engine-related heuristics. That evidence is intentionally weak: it shows that the software seam is wired and that the procedural prior is not condition-invariant, but it does not validate mission-conditioned or manufacturing-conditioned aircraft generation on grounded aircraft-like data.

F. Internal Solver Validation against Canonical Benchmarks

To verify the physical realism of the internal D3Q27 solver, we compare its Drag Coefficient (C_d) results against canonical benchmarks for two standard geometries: a circular cylinder and a unit cube. These tests use the internal solver's momentum-exchange method with BFL boundary conditions.

TABLE II
INTERNAL SOLVER C_d SANITY TARGETS AND OBSERVATIONS

Geometry	**Reynolds No.**	**Internal Solver C_d**	**Literature Target (C_d)**
Circular Cylinder	100	1.56	1.3 – 1.4 [25]
Unit Cube	1000	1.04	$\sim$1.05 [26]

The results in Table II provide an informal sanity check for our LBM implementation, framing the observations against typical literature targets for 3D flow. The tests were conducted in a $10L \times 5L \times 5L$ domain (L: characteristic length) using a 128^3 grid, no-slip BFL wall conditions, and a Neumann outlet. We mapped physical velocity to $Ma = 0.025$ ($u_{lattice} \approx 0.014$) and averaged forces over the final 125 steps of a 500-step simulation to ensure stability.

Following physics corrections to the freestream lattice velocity initialization and BFL momentum exchange weighting ($4.0/(1 + 2q)$), the solver yields C_d values that align with the broad expectations of the literature. While exact 3D benchmarks are sensitive to blockage ratios and averaging criteria, these results confirm the solver's utility for providing simulation-in-the-loop guidance. These calibrated physics enable the generation of aerodynamic labels for surrogate training, though rigorous validation remains tiered behind high-fidelity PDE solvers.

G. Discussion

The main takeaway is now a little sharper: the repository's end-to-end path executes, the shape prior is wired into training, and the export path emits a concrete OpenFOAM validation case with the generated STL in `constant/triSurface/design.stl`. On the same reduced cube benchmark, the internal D3Q27 solver and the OpenFOAM sonicFoam run both complete on the shared validation geometry, providing a direct solver-to-solver comparison for the current pipeline. The result supports a reproducible implementation path, not a claim of aircraft-level aerodynamic optimization.

The OpenFOAM export path now writes a plain `forces` function-object dictionary, and the resulting force vector is normalized manually to recover coefficients when coefficient output is not available. The current benchmark still shows a substantial Cd gap, which we are now treating as a solver- and sampling-consistency issue rather than a paper-level normalization issue. A higher-confidence CFD comparison will still need a more systematic hyperparameter sweep and consistent normalization of reference area and flow conditions.

The next recorded physics correction was applied in `CLI/cascaded_lbm.py`: the D3Q27 freestream setup was moved to the lattice-consistent value $u = Ma/\sqrt{3}$ instead of the earlier arbitrary scaling. That change is now part of the benchmark history and is being checked against the Cd mismatch before any deeper transform rewrite.

V. Validation and Testing Standards

To make the validation story honest, we separate three levels of evidence:

- **Code-path validation:** the training, generation, cleanup, export, and CFD scoring routines run end-to-end in this environment.
- **Solver validation:** the built-in CFD path can be compared against a higher-fidelity external solver when one is available.
- **Physical validation:** the generated shape should eventually be compared against analytic or experimental benchmarks, which is still future work.

A. Verification

Verification checks that the implemented steps are executed correctly. In this revision we verified that:

- the shape-plausibility loss contributes to training,
- the connected-component cleanup hook reduces fragmented voxel outputs,
- STL export succeeds from the cleaned voxel grid, and
- the OpenFOAM export hook writes a runnable case directory skeleton, including ‘blockMeshDict‘, ‘snappyHexMeshDict‘, and surface STL output.

B. Validation

Validation is now reproducible on a single local machine with OpenFOAM installed. The built-in CFD path is exercised on the reduced CPU pipeline, and the OpenFOAM sonicFoam benchmark is run on the shared centered-cube validation object. The comparison uses the lower-level ‘forces‘ output when available, with a manual pressure integration fallback only if the function-object output is missing.

The paper therefore claims the following and nothing more: the export path is concrete, the benchmark comparison is measurable, and the shared validation geometry is documented explicitly so the solver-to-solver check can be repeated without ambiguity.

VI. Conclusion

The present implementation demonstrates a working proof-of-concept generative-design pipeline and a reduced freeform-object experiment that can be executed in the installed environment. The observed step sweep suggests that longer consistency sampling can change the reported L/D in this specific sanity setting, but those numbers remain tied to downsampled freeform objects and CPU CFD evaluation.

The OpenFOAM cross-check completes on the shared centered-cube validation object and gives the repository a reproducible external CFD reference path. The next revision should therefore prioritize solver calibration and a larger ablation study before presenting stronger aerodynamic conclusions.

Most importantly, the repository should not yet be described as a fully AI-driven airplane generator conditioned on flight profile, manufacturing method, or structural requirements. The current branch does contain partial structured-conditioning plumbing and smoke-level condition-response artifacts, but reaching a claim-bearing conditioned generator would still require a real aircraft-like dataset, explicit conditional evaluation, stronger structural and aerodynamic gates, and hardware or cloud runs beyond the present sanity-run budget.

References

[1] A. Elham and M. J. L. van Tooren, “Discrete adjoint aerodynamic shape optimization using symbolic analysis with openfemflow,” *Structural and Multidisciplinary Optimization*, vol. 63, pp. 2531–2551, 2021.

[2] E. Andrés-Pérez and C. Paulete-Periáñez, “On the application of surrogate regression models for aerodynamic coefficient prediction,” *Complex & Intelligent Systems*, vol. 7, pp. 1991–2021, 2021.

[3] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 6840–6851, 2020.

[4] A. Q. Nichol and P. Dhariwal, “Improved denoising diffusion probabilistic models,” in *Proceedings of the 38th International Conference on Machine Learning*, vol. 139. PMLR, 2021, pp. 8162–8171.

[5] A. Nichol, H. Jun, P. Dhariwal, P. Mishkin, and M. Chen, “Point-e: A system for generating 3d point clouds from complex prompts,” *arXiv preprint arXiv:2212.08751*, 2022.

[6] H. Jun and A. Nichol, “Shap-e: Generating conditional 3d implicit functions,” *arXiv preprint arXiv:2305.02463*, 2023.

[7] J. Wu, O. Sigmund, and J. P. Groen, “Topology optimization of multi-scale structures: a review,” *Structural and Multidisciplinary Optimization*, vol. 63, pp. 1455–1480, 2021.

[8] Y. Li, Y. Sun, P. Ma, E. Sifakis, T. Du, B. Zhu, and W. Matusik, “Neuralfluid: Neural fluidic system design and control with differentiable simulation,” *arXiv preprint arXiv:2405.14903*, 2024.

[9] O. Sigmund, “A 99 line topology optimization code written in matlab,” *Structural and Multidisciplinary Optimization*, vol. 21, pp. 120–127, 2001.

[10] M. P. Bendsøe and O. Sigmund, *Topology Optimization: Theory, Methods, and Applications*. Springer Science & Business Media, 2013.

[11] I. Sosnovik and I. Oseledets, “Neural networks for topology optimization,” *Structural and Multidisciplinary Optimization*, vol. 60, no. 5, pp. 1387–1401, 2019.

[12] A. Jameson, “Aerodynamic design via control theory,” *Journal of Scientific Computing*, vol. 3, pp. 233–260, 1988.

[13] ——, “Optimum aerodynamic design via boundary control,” in *12th Computational Fluid Dynamics Conference*, 1995, p. 2432.

[14] A. Forrester, A. Sobester, and A. Keane, *Engineering design via surrogate modelling: a practical guide*. John Wiley & Sons, 2008.

[15] M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[16] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar, “Fourier neural operator for parametric partial differential equations,” *arXiv preprint arXiv:2010.08895*, 2020.

[17] L. Lu, P. Jin, G. Pang, and G. E. Karniadakis, “Learning nonlinear operators via deeponet based on the universal approximation theorem of operators,” *Nature Machine Intelligence*, vol. 3, no. 3, pp. 218–229, 2021.

[18] R. Vinuesa and S. L. Brunton, “Enhancing computational fluid dynamics with machine learning,” *Nature Computational Science*, vol. 2, no. 6, pp. 358–366, 2022.

[19] Z. Wang, K. Zhang, J. Chu, and H. Ma, “Physics-informed machine learning for aero-structural design optimization,” *Progress in Aerospace Sciences*, vol. 136, p. 100868, 2023.

[20] S. Oh, Y. Jung, S. Kim, I. Lee, and N. Kang, “Deep learning-based generative design for skeleton-optimized structures,” *Engineering with Computers*, vol. 35, pp. 1279–1293, 2019.

[21] F. Steinmetz, S. J. Pfertner, and N. R. Gauger, “Generative design of manufacturable structures using differentiable projection,” *Computer-Aided Design*, vol. 145, p. 103168, 2022.

[22] J. Wu, C. Zhang, T. Xue, W. T. Freeman, and J. B. Tenenbaum, “Learning a probabilistic latent space of object shapes via 3d generative-adversarial modeling,” in *Advances in Neural Information Processing Systems*, vol. 29, 2016.

[23] S. Luo and W. Hu, “Diffusion probabilistic models for 3d point cloud generation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 2837–2845.

[24] L. Mescheder, M. Oechsle, M. Niemeyer, S. Nowozin, and A. Geiger, "Occupancy networks: Learning 3d reconstruction in function space," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 4460–4470.

[25] M. Schäfer, S. Turek, F. Durst, E. Krause, and E. Rank, "Benchmark computations of laminar flow around a cylinder," in *Flow Simulation with High-Performance Computers II*. Springer, 1996, pp. 547–566.

[26] M. Geier, A. Pasquali, and M. Schonherr, "The cascaded lattice boltzmann method," *Journal of Computational Physics*, vol. 300, pp. 516–542, 2015.